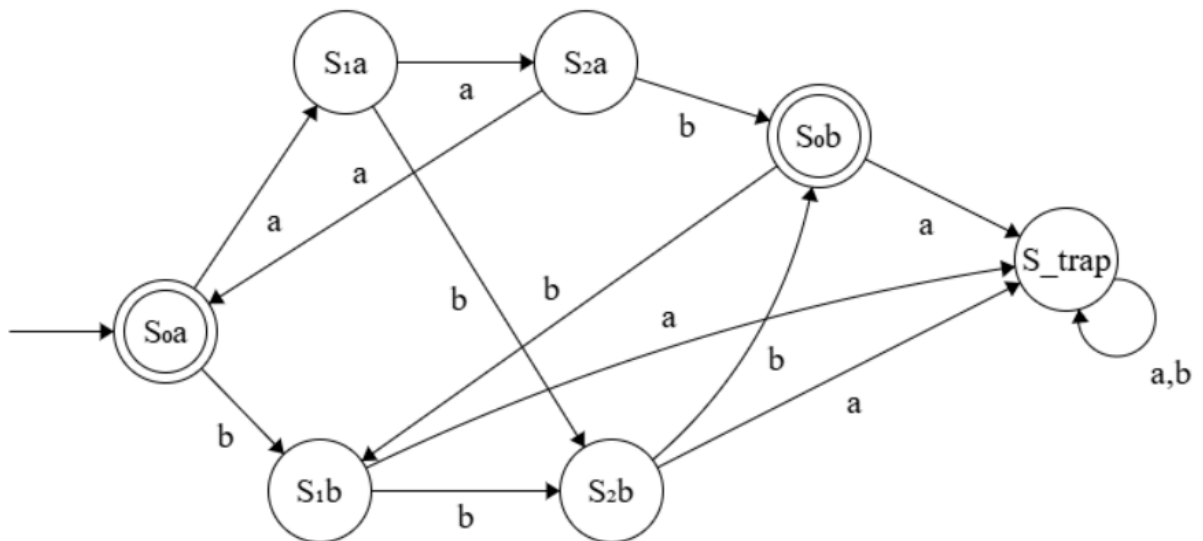
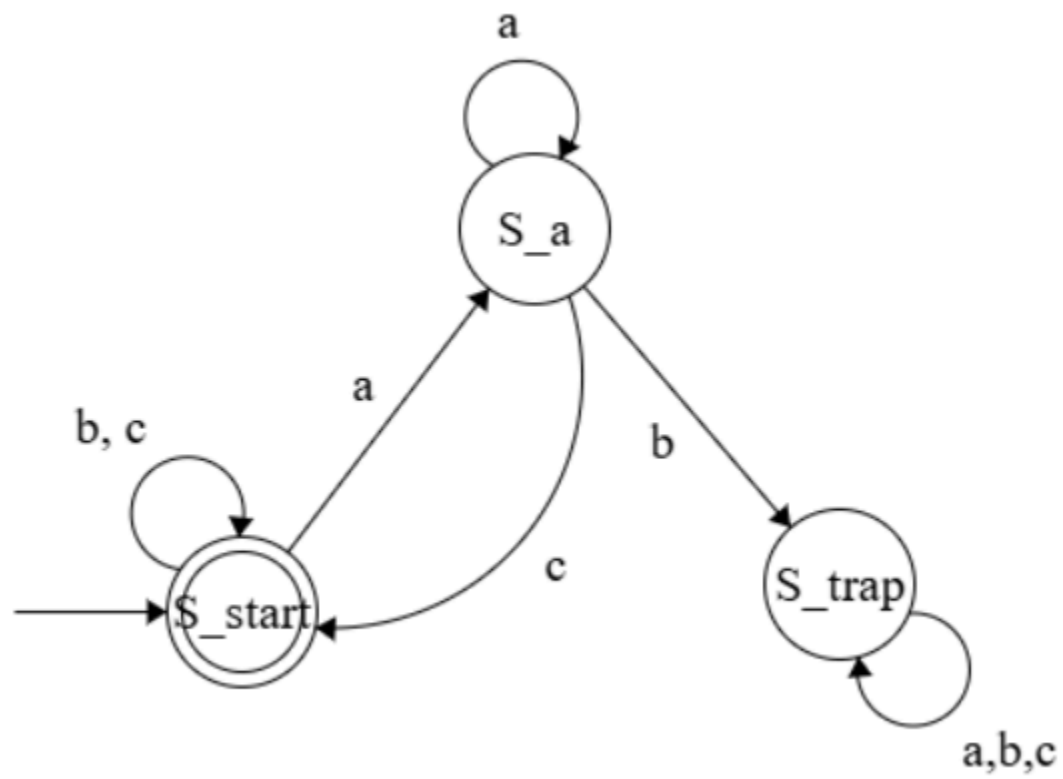


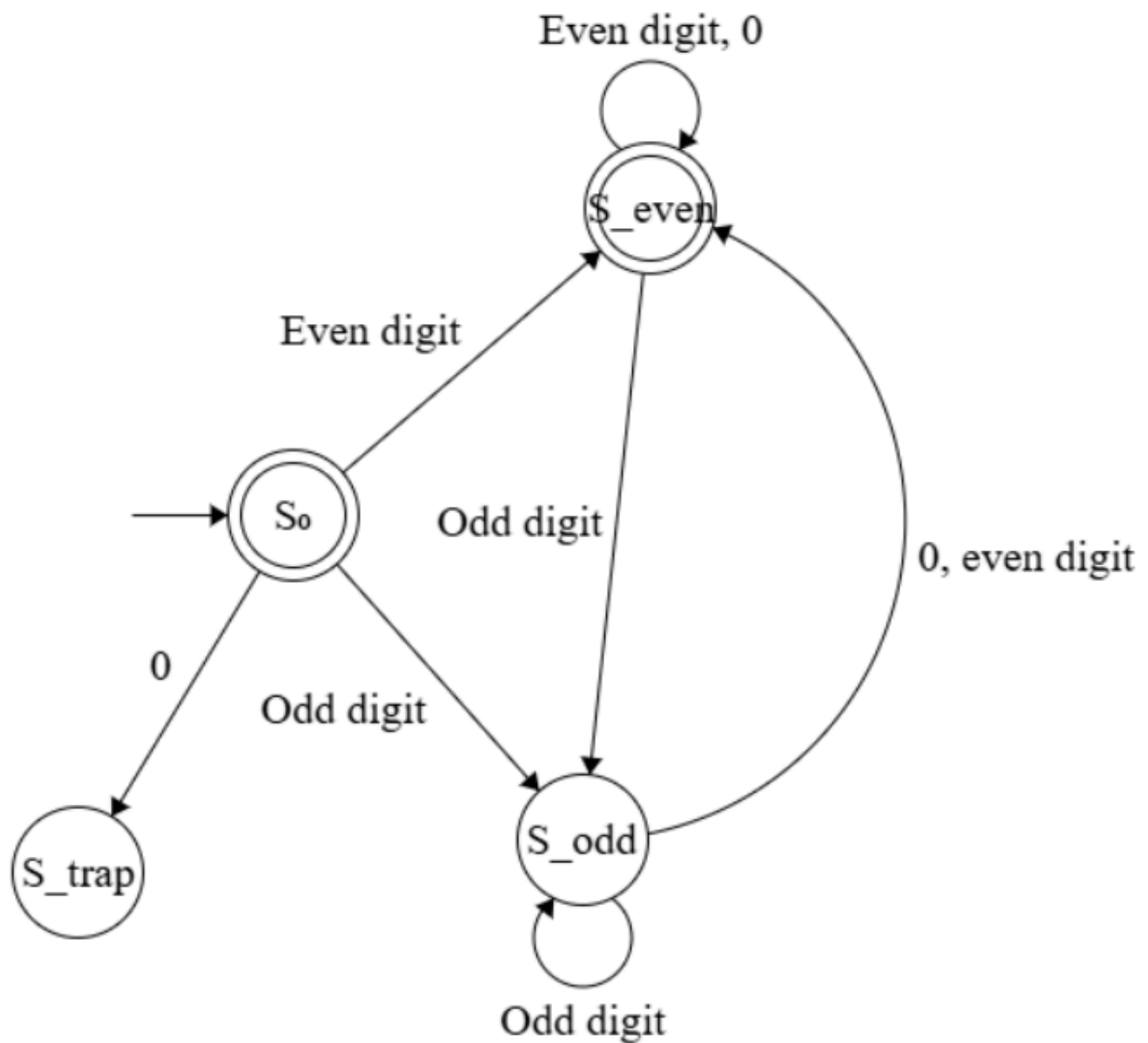
1.



2.



3.



4.

NFA Graph Description:

- Start at state A
- If 0, then go to top route and to state b.
 - At state b, loop on 0 and 1
 - If 0, then go to state c
 - If 0, then go to state d
 - If 0, then go to state h
- If 1, then go to bottom route and to state e
 - At state e, loop on 0 and 1
 - If 1, then go to state f
 - If 1, then go to state g
 - If 1, then go to state h

- At state h, loop on 0 and 1
- Accept on state h

Language Description:

$\Sigma = \{0, 1\} \rightarrow$ Language consists of strings with 0's and 1's.

Language contains all binary strings with either at least three consecutive 0's or three consecutive 1's anywhere in the string. The strings otherwise may have any combination and any amount of 0's and 1's.

PRECISE:

Set of all strings constructed with any combination of 0's and 1's, followed by either 000 or 111, followed then by any combination of 0's and 1's.

Regular Expression:

$(0^*1^*)^*(000 + 111)(0^*1^*)^*$

$\subseteq \wedge \in \forall \exists \neg \Sigma \varepsilon$

5.

